UNIT

Two

Behavior of the function and curve sketching

Exercise 5 Increasing and decreasing functions.

Exercise 6 Local maxima and minima (Local extrema).

Exercise 7 Absolute maxima and minima (Absolute extrema).

Exercise 8 Convexity of curves and inflection points.

Exercise 9 Curve studying.

Exercise 10 Applications of maxima and minima.



Exercise 5

On increasing and decreasing functions

From the school book | Understand | Apply | Higher Order Thinking Skills

1 Choose the correct answer from those given :

(1) The first derivative of the decreasing functions are ..... for all values of domain.

(a) positive (b) negative (c) zero (d) otherwise

(2) If f is a continuous function on $\mathbb{R}$ and the function f has a critical point at $\mathcal{X} = a$, so .....

(a) $\hat{f}(a) = 0$ (b) $\hat{f}(a)$ is undefined.

(c) $\hat{f}(a^+) \neq \hat{f}(a^-)$ (d) All the previous.

(3) The function $f : f(\mathcal{X}) = -\mathcal{X}^2$ is increasing in the interval .....

(a) $]0 , \infty[$ (b) $]-\infty , 0[$ (c) $\mathbb{R} - \{0\}$ (d) $\mathbb{R}$

(4) The function $f : f(\mathcal{X}) = -|\mathcal{X}| + 1$ is decreasing in the interval .....

(a) $]0 , \infty[$ (b) $]-\infty , 1[$ (c) $]-1 , \infty[$ (d) $]-\infty , 0[$

(5) The function $f : f(\mathcal{X}) = \mathcal{X}^3 + 4\mathcal{X} + 2$ is increasing when $\mathcal{X} \in$ .....

(a) $]-4 , \infty[$ (b) $\mathbb{R}$ (c) $]-\infty , \frac{-4}{3}[$ (d) $]\frac{-4}{3} , \infty[$

(6) *(2nd Session 2014)* The function $f : f(\mathcal{X}) = (\mathcal{X} - 3)^2 + 2$ is increasing in the interval .....

(a) $]2 , \infty[$ (b) $]3 , \infty[$ (c) $]-\infty , 3[$ (d) $]0 , \infty[$

(7) *(1st Session 2022)* If the function $f : f(\mathcal{X}) = \mathcal{X}^3 - 6\mathcal{X}^2 + 15$, then the function f is decreasing in the interval .....

(a) $]0 , 4[$ (b) $]4 , \infty[$ (c) $]-\infty , 0[$ (d) $\mathbb{R} -]0 , 4[$

(8) *(2nd Session 2023)* The function $f : f(\mathcal{X}) = |\mathcal{X} - 3|$ is .....

(a) decreasing in the interval $]-\infty , 3[$ (b) decreasing in the interval $]-3 , \infty[$

(c) increasing in the interval $]-\infty , 3[$ (d) increasing in the interval $]-3 , \infty[$

(9) *(Speciman 2025)* If the curve $y = (a - 2\mathcal{X})^2$ has a critical point at $\mathcal{X} = -1$, then a = ...

(a) 2 (b) – 1 (c) – 2 (d) 1

(10) If the function $f : f(x) = x^2 + \frac{b}{x}$ has a critical point at $x = 2$, then b = ...

(a) – 16 (b) 16 (c) – 4 (d) 2

(11) If $f : f(x) = a\,x^2 + b\,x + 2$ has a critical point (1 , 4) , then a – b = ...

(a) 2 (b) zero (c) – 6 (d) – 8

(12) All the following functions are increasing in their domain except the function $f : f(x) =$ .....

(a) $2x - 17$ (b) e^x (c) $(x-3)^2$ (d) x^3

(13) If $f : f(x) = \sqrt[3]{x-c}$ and it has a critical point at (c , 0) , then $\hat{f}(c)$ .....

(a) undefined. (b) zero. (c) $\frac{1}{3}$ (d) $\frac{1}{3\sqrt[3]{c^2}}$

(14) If $f(x) = x \ln x$, then the function $f : f(x)$ has a critical point at $x =$ .....

(a) zero (b) 1 (c) e (d) $\frac{1}{e}$

(15) *(2nd Session 2022)* If the function $f : f(x) = 3 - 2\,e^{3x^2}$, then the function is decreasing in the interval .....

(a) $]-\infty , 12[$ (b) $]-12 , 12[$ (c) $]0 , \infty[$ (d) $]-\infty , 0[$

(16) *(Speciman 2025)* If $f(x) = e^{8x - x^2}$, then the function is increasing in the interval .....

(a) $]-\infty , 0[$ (b) $]0 , \infty[$ (c) $]4 , \infty[$ (d) $]-\infty , 4[$

(17) If the function f is a polynomial function of fourth degree and its domain is $\mathbb{R}$, then the greatest number of the critical points of the function $f(x)$ is .....

(a) 1 (b) 2 (c) 3 (d) 4

(18) *(2nd Session 2017)* If $f :]-1 , 4[\longrightarrow \mathbb{R}$, $f(x) = x^3 - 3x$, then the number of the critical points of the function f equals .....

(a) zero (b) 1 (c) 2 (d) 3

(19) *(1st Session 2018)* If $f(x) = x(a - \ln x)$ where a is constant and the curve of the function has a critical point at $x = e$, then a = ...

(a) 1 (b) zero (c) e (d) 2

(20) The function $f : f(x) = \frac{x}{\ln x}$ is increasing in the interval .....

(a) $]0 , \infty[$ (b) $]0 , e[$ (c) $]e , \infty[$ (d) $]\frac{1}{e} , e^2[$

(21) *(1st Session 2019)* If f is a function where $f(x) = \frac{x^4 + 1}{x^2}$, then the function is decreasing in .....

(a) $]-\infty , -1[$ only. (b) $]0 , 1[$ only.

(c) $]-1 , 0[,]1 , \infty[$ (d) $]-\infty , -1[,]0 , 1[$

(22) *(2nd Session 2021)* If the function $f(x) = x^2 + \frac{2a^3}{x}$ where $a \in \mathbb{R}^-$, then the function is decreasing in the interval .....

(a) $]-\infty , 0[,]0 , \infty[$ (b) $]a , \infty[$

(c) $]-\infty , a[$ (d) $]-\infty , 0[,]a , 0[$

(23) *(2nd Session 2019)* If f is function where $f(x) = (x^2 - 4)^{\frac{2}{3}}$, then the function is decreasing in .....

(a) $]-\infty , -2[,]0 , 2[$ (b) $]-2 , 0[,]2 , \infty[$

(c) $]-\infty , -2[$ only. (d) $]0 , 2[$ only.

(24) *(Speciman 2025)* The function $f : f(x) = \frac{1}{3}x + \sqrt[3]{x^2}$ is decreasing in the interval .....

(a) $]0 , 8[$ (b) $]-8 , \infty[$ (c) $]-8 , 0[$ (d) $\mathbb{R} -]-8 , 0[$

(25) *(Speciman 2025)* The function $f : f(x) = \ln(x^2 - 4)$ is increasing in the interval .....

(a) $]0 , 8[$ (b) $]-\infty , 0[$ (c) $]2 , \infty[$ (d) $]-\infty , 2[$

(26) *(2nd Session 2023)* If $f(x) = \ln(x^2 + 1)^3 - 2$, then the function f has a critical point at $x =$ .....

(a) e^2 (b) -1 (c) zero (d) 1

(27) *(1st Session 2022)* If $f : f(x) = \sqrt[3]{x^2 - 6x}$, then the number of the critical points for the curve of the function f is .....

(a) zero (b) 1 (c) 2 (d) 3

(28) *(Speciman 2025)* The function $f : f(x) = x - \ln x^2$ is decreasing in the interval .....

(a) $]2 , \infty[$ (b) $]-\infty , 0[$ (c) $]0 , 2[$ (d) $\mathbb{R}$

(29) *(Speciman 2025)* When studying the increasing and decreasing intervals of the function $f : [\frac{1}{e} , e] \longrightarrow \mathbb{R}$, $f(x) = x - \ln x$, then the function is .....

(a) decreasing in $]\frac{1}{e} , 1[$ and increasing in $]1 , e[$

(b) increasing in $]\frac{1}{e} , 1[$ and decreasing in $]1 , e[$

(c) decreasing in the two intervals $]\frac{1}{e} , 1[,]1 , e[$

(d) increasing in the two intervals $]\frac{1}{e} , 1[,]1 , e[$

2 **Find the critical points then determine the increasing and decreasing intervals of the following functions :**

(1) $y = x^2 - 6x + 7$

(2) $y = 1 - 4x - x^2$

(3) $f(x) = x^3 - 12x + 7$

(4) 📖 $f(x) = x^3 - 6x^2 + 5$

(5) $y = (x-2)^2$

(6) $f(x) = (x+4)^3$

(7) $f(x) = x^3 - 9x^2 + 24x - 4$

(8) 📖 $f(x) = 9x - x^3$

(9) *(1st Session 2005)* $f(x) = x(x-3)^2$

(10) 📖 $f(x) = x^4 + 4x$

(11) $f(x) = x^4 - 8x^2 + 16$

(12) $f(x) = x^5 - 5x^4 + 1$

(13) $f(x) = (x^2 - 9)^2$

(14) $f(x) = x^3 (x-1)^2$

(15) $f(x) = (x+2)(x-1)^2$

(16) $f(x) = 5x^6 - 6x^5 + 1$

(17) $f(x) = x^4 + 2x^3 - 3x^2 - 4x$

3 **Find the critical points then determine the increasing and decreasing intervals of each of the following function :**

(1) $f(x) = \frac{x}{x+2}$

(2) $f(x) = \frac{x^2}{x-2}$

(3) $f(x) = \frac{1}{x-3}$

(4) $f(x) = \sqrt{2x-3}$

(5) $f(x) = \sqrt{4-x}$

(6) $f(x) = \sqrt[3]{x-1}$

(7) $f(x) = \sqrt[3]{2-x}$

(8) $y = \frac{1}{3}x - \sqrt[3]{x}$

(9) $f(x) = x^{\frac{2}{3}}$

(10) $f(x) = \sqrt[3]{(x-2)^2}$

(11) $f(x) = \sqrt[3]{(x^2-1)^2}$

(12) 📖 $f(x) = 2 - 3(x-2)^{\frac{4}{3}}$

(13) 📖 $f(x) = 1 - \frac{1}{x}$

(14) $f(x) = x + \frac{1}{x}$

(15) $f(x) = \sqrt{x} - \frac{1}{\sqrt{x}}$

(16) $f(x) = \frac{x^2 + x - 2}{x^2 - 1}$

(17) 📖 $f(x) = \frac{x}{x^2+1}$

(18) $f(x) = \sqrt{9-x^2}$

(19) $f(x) = \sqrt{x^2 - 4x + 3}$

(20) $f(x) = |x-3| + 2$

(21) $f(x) = 3 - |x-2|$

(22) $f(x) = 2 - \sqrt{x^2 - 6x + 9}$

(23) $f(x) = x|x|$

(24) $f(x) = x^2|x|$

(25) $f(x) = x|x-2|$

(26) $f(x) = \frac{\sqrt{x-1}}{x}$

(27) $f(x) = |x^2 - 5x + 6|$

(28) $f(x) = x^2\sqrt{7-x^2}$

4 Find the critical points and determine the increasing and decreasing intervals of the following functions in the given domain :

(1) $f(x) = 1 - \sin x$, $0 < x < 2\pi$

(2) $f(x) = x + 2\sin x$, $0 < x < 2\pi$

(3) $f(x) = x - 2\cos x$, on $[0, 2\pi[$

(4) $f(x) = \sin 2x$ on $[0, \pi[$

(5) $f(x) = \sin x + \sin^2 x$ on $[0, 2\pi[$

(6) $f(x) = 4x - \tan x$ on $[0, \frac{\pi}{2}[$

5 Choose the correct answer from the given ones :

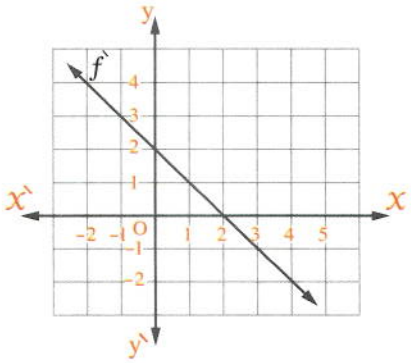
(1) *(2nd Session 2024)* If $\grave{f}(x) = 2x - 8$, then the function f is increasing on the interval .....

(a) $]-\infty, 4[$ (b) $]4, \infty[$ (c) $]-\infty, \infty[$ (d) $]-4, 4[$

(2) If $f'(x) = (x-1)(x-3)$, then the function f is decreasing in the interval .....

(a) $]1, 3[$ (b) $]2, \infty[$ (c) $]-\infty, 2[$ (d) $\mathbb{R} - [1, 3]$

(3) The opposite figure represents f' , then the function f is increasing in the interval .....



(a) $]0, \infty[$ (b) $]-\infty, 0[$

(c) $]2, \infty[$ (d) $]-\infty, 2[$

(4) If $f : \mathbb{R} \longrightarrow \mathbb{R}$ where $f(x) = x^3 - 27x + 5$, then the function f is increasing when .....

(a) $x < -3$ (b) $|x| < 3$ (c) $x \leq -3$ (d) $|x| > 3$

(5) If the function $f : f(x) = 2 - ax^3$ is decreasing on its domain , then .....

(a) $a \leq 0$ (b) $a > 0$ (c) $a \geq 0$ (d) $a = -\frac{1}{2}$

(6) If $f : \mathbb{R}^* \longrightarrow \mathbb{R}$ where $f(x) = \ln|x| + ax^2 + bx$ has two critical points at $x = -1$ and $x = 2$, then $a + b =$ .....

(a) $\frac{-1}{4}$ (b) $\frac{1}{4}$ (c) $\frac{1}{2}$ (d) $\frac{-1}{2}$

(7) If g is increasing function on $\mathbb{R}$, h is decreasing function on $\mathbb{R}$ and $f(x) = 4g(x) - 3h(x)$, then function f is ..... on $\mathbb{R}$

(a) increasing (b) decreasing (c) constant (d) zero

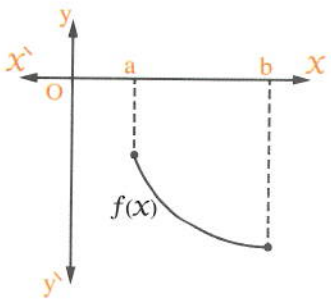
(8) f , g are two increasing functions on $\mathbb{R}$, which of the following is increasing on its domain ?

(a) $f + g$ (b) $f - g$ (c) $f \cdot g$ (d) $\frac{f}{g}$

(9) Let f be an increasing function on its domain , which of the following function not necessary to be increasing on its domain ?

(a) $y = \sqrt{f(x)}$ (b) $y = \sqrt[3]{f(x)}$ (c) $y = [f(x)]^2$ (d) $y = e^{f(x)}$

(10) The opposite figure represents the curve of function f , its domain is $[a , b]$, then the function $g : g(x) = x \cdot f(x)$ is ..... in the interval $]a , b[$



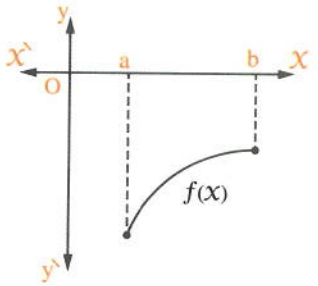
(a) decreasing

(b) increasing

(c) constant

(d) not possible to determine its monotony

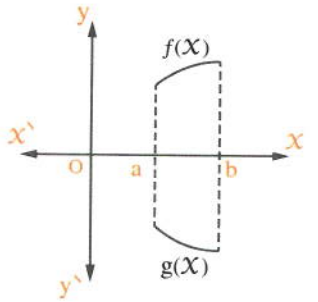
(11) The opposite figure represents the curve of function f where $f : [a , b] \longrightarrow \mathbb{R}^-$, then the function $g : g(x) =$ ..... is increasing in the interval $]a , b[$



(a) $[f(x)]^2$ (b) $x \times f(x)$

(c) $[f(x)]^3$ (d) $-2x - f(x)$

(12) *(Speciman 2025)* If $y = f(x)$, $y = g(x)$ are two differentiable functions in the interval $[a , b]$, which of the following functions is always increasing in the interval $[a , b]$?

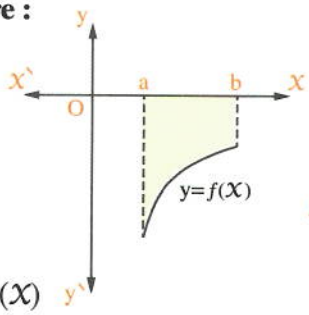


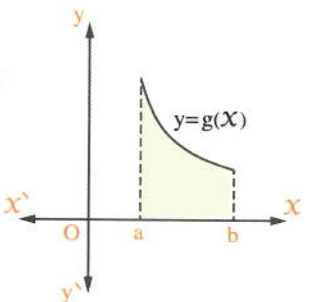
(a) $f(x) + g(x)$ (b) $f(x) - g(x)$

(c) $f(x) \cdot g(x)$ (d) $\frac{f(x)}{g(x)}$

(13) *(Speciman 2025)* **In the opposite figure :**

All the following functions are always increasing in the interval $]a , b[$ except .....





(a) $f(x) - g(x)$ (b) $[f(x)]^3$

(c) $[g(x)]^2$ (d) $f(x) \cdot g(x)$

(14) If the function $f : f(x) = \frac{x}{x^2+1}$ is increasing on $]a , b[$, then the greatest of the expression : $(b - a) =$ .....

(a) zero (b) 1 (c) 2 (d) 3

(15) If $f : \mathbb{R} \longrightarrow \mathbb{R}$ where $f(x) = \sqrt[3]{x^2}\,(3x - 7)$ is increasing for every $x \in]-\infty , a[$, $x \in]b , \infty[$, then $7a + 15b =$ .....

(a) 7 (b) 14 (c) 21 (d) 22

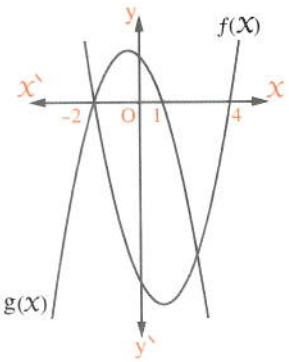
(16) If the function f is defined in $]3 , 8[$ where $f(x_1) > f(x_2)$ for every $x_2 > x_1$ and the function g is defined on $]-3 , 2[$ where $g(x_1) - g(x_2) > 0$ for every $x_1 > x_2$, then all the following statements are true except .....

(a) $\dot{f}(4) < \dot{g}(-2)$ (b) $\frac{f(5) - f(7)}{\dot{g}(0)} > 0$

(c) $\dot{f}(6) > \dot{g}(-1)$ (d) $\frac{f(6) - f(4)}{\dot{g}(-1)} < 0$

(17) The opposite figure represents the graphs of f and g

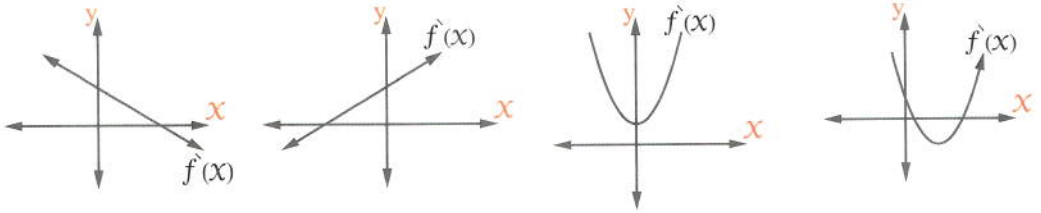
If $\dot{h}(x) = \dot{f}(x) \cdot g(x)$, then h is increasing on .....



(a) $]-\infty , -2[$ (b) $]-2 , 1[$

(c) $]1 , 4[$ (d) $]4 , \infty[$

(18) If $f : \mathbb{R} \longrightarrow \mathbb{R}$ and for all values of $x \in \mathbb{R}$, f is increasing function , then the figure which represents $\dot{f}(x)$ is .....



(a) (b) (c) (d)

6 Prove that : f where $f(x) = \tan x - x$ is increasing on the interval $]0 , \frac{\pi}{4}[$

7 If f , g are two differentiable functions , $\dot{f}(x) < \dot{g}(x)$ when $x \in \mathbb{R}$, prove that : the function $h(x) = f(x) - g(x)$ is decreasing when $x \in \mathbb{R}$

8 **Determine the increasing and decreasing intervals of the function f in each of the following :**

(1) $f(x) = x - e^{x}$

(2) $f(x) = 5 - 2\,e^{2x}$

(3) $f(x) = x + \ln x$

(4) $f(x) = 2x - \ln x$

(5) $f(x) = 3 - \ln x^{2}$

(6) $g(x) = 2 \ln x - x^{2}$

9 **Find the critical points then determine the increasing and decreasing intervals of the following function :**

(1) $f(x) = \begin{cases} x^{2} + 2 & , \ x \leq 1 \\ 2x + 1 & , \ x > 1 \end{cases}$

(2) $f(x) = \begin{cases} 7 - 2x & , \ x < 3 \\ 3x - 8 & , \ x \geq 3 \end{cases}$

(3) $f(x) = \begin{cases} 3 - 4x - x^{2} & , \ x \leq 1 \\ 2x - 4 & , \ x > 1 \end{cases}$

(4) $f(x) = \begin{cases} 3x + 1 & , \ x \leq 1 \\ 5 - x^{2} & , \ x > 1 \end{cases}$

(5) $f(x) = \begin{cases} 3x + 5 & , \ x < -1 \\ x^{2} + 1 & , \ -1 \leq x < 2 \\ 9 - x^{2} & , \ x \geq 2 \end{cases}$

10 **Find the value of each for a , b , c and d where the curve of $f(x) = a\,x^{3} + b\,x^{2} + c\,x + d$ satisfies the following conditions together :**

(a) It passes through the origin point.

(b) It has a critical point when $x = 1$

(c) The equation of the tangent to the curve at point $(2 , f(2))$ on it is $9x + y = 20$

« 1 , − 9 , 15 , 0 »

11 **Determine the increasing and decreasing intervals in each of the following :**

(1) $f(x) = \begin{cases} 3 - (x-2)^{2} & , \ -4 \leq x \leq 2 \\ 3 & , \ 2 < x < 4 \\ 7 - x & , \ 4 \leq x \leq 8 \end{cases}$

(2) $f(x) = \begin{cases} 2x - x^{2} & , \ x \leq 1 \\ 1 & , \ 1 < x \leq 4 \\ x^{2} - 8x + 17 & , \ x > 4 \end{cases}$

Higher skills

12 **Choose the correct answer from those given :**

(1) If f is a differentiable function on $\mathbb{R}$ such that $f'(x) < 0$ for all values of $x \in \mathbb{R}$, then .....

(a) $f(x) > f(x-1)$

(b) $f(x) < f(x+1)$

(c) $f(x) < f(x-1)$

(d) $f(x) + f(x+1) = 1$

(2) If f is a polynomial function and $f'(x) = a x^2 + b x + c$, then f is decreasing in its domain if .....

(a) $a > 0 , b^2 - 4ac \leq 0$ (b) $a > 0 , b^2 - 4ac \geq 0$

(c) $a < 0 , b^2 - 4ac \leq 0$ (d) $a < 0 , b^2 - 4ac \geq 0$

(3) If $f(x) = x^3 + a x^2 + 3x - 2$ is increasing on $\mathbb{R}$ = then $a \in$ .....

(a) $]3 , \infty[$ (b) $[-3 , 3]$ (c) $]-\infty , 3[$ (d) $\mathbb{R}$

(4) If the function $f : f(x) = k x^3 + 5x$ has two critical points , then $k \in$ .....

(a) $]0 , \infty[$ (b) $[0 , \infty[$ (c) $]-\infty , 0[$ (d) $]-\infty , 0]$

(5) If f is a positive increasing functions , g is a positive decreasing functions and $z(x) = \frac{f(x)}{g(x)}$, then z is .....

(a) negative and increasing. (b) negative and decreasing.

(c) positive and increasing. (d) positive and decreasing.

(6) If function $f : \mathbb{R} - \{1\} \longrightarrow \mathbb{R}$ where $f(x) = \frac{x^2 + m x + 4}{x - 1}$, where m is a constant is an increasing function , then $m \in$ .....

(a) $]-\infty , -5]$ (b) $[-5 , \infty[$

(c) $]-7 , 0]$ (d) $]-10 , 10[$

(7) If $f : \mathbb{R} \longrightarrow \mathbb{R}$ where $f(x) = x^3 + a x^2 + 12x + 1$, then the set of values of a which makes the function f has no critical point is .....

(a) $a < -6$ (b) $-6 < a < 6$ (c) $a > 6$ (d) $-3 < a < 9$

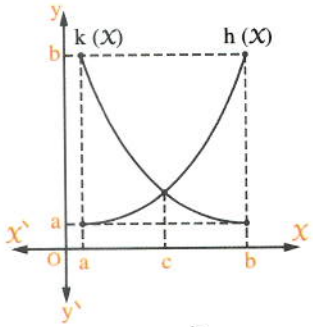
(8) *(Trial 2021)* The opposite figure represents the curves of the two functions h and k in the interval $[a , b]$ if f is a function where $f(x) = (h \circ k)(x)$, then the correct statement from the following is .....

(a) The function f is decreasing on $]a , b[$

(b) The function f is increasing on $]a , b[$

(c) The function f is increasing on $]a , c[$ only.

(d) The function f is decreasing on $]a , c[$ only.



(9) *(Trial 2021)* If function g is differentiable twice in the interval $]0 , 1[$ and $\ddot{g}(x) < 0$ and if f is a function where $f(x) = g(x) + g(1 - x)$, then the correct statement from the following is .....

(a) f is decreasing on $]\frac{1}{2} , 1[$ (b) f is increasing on $]\frac{1}{2} , 1[$

(c) f is decreasing on $]0 , 1[$ (d) f is increasing on $]0 , 1[$